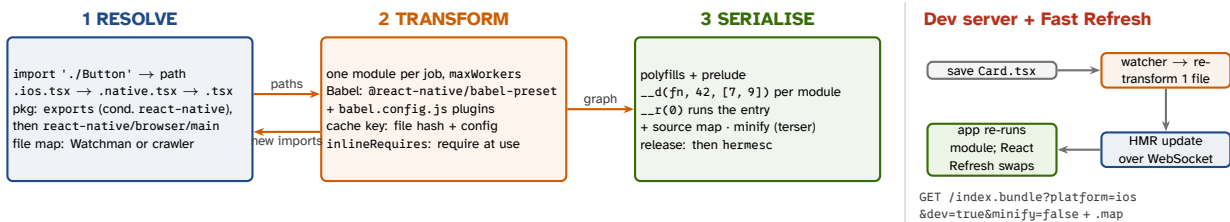


In one line: Metro turns an entry file into **one JS bundle + source map** in three stages: **resolve** every import into a dependency graph, **transform** each module with Babel in parallel workers (cached per file), **serialise** the modules as `__d(factory, id, deps)` definitions plus polyfills and a `__r(entry)` call. In development it is also the **server on :8081**, pushing changed modules over a WebSocket for **Fast Refresh**.



How it works

- Resolution and transformation overlap:** each transformed module reveals new imports to resolve. sourceExts default `js jsx json ts tsx`; anything in `assetExts` becomes an asset record (scales @2x/@3x), not code.
- Package exports/imports** resolution is on by default since Metro 0.82 (RN 0.79); conditions: `react-native` (+ `browser` on web), `require/import`.
- Transform cache** (`cacheStores`, default files in `os.tmpdir()`) is keyed by file content and transform options — a change in Babel config or an *inlined env var* can leave stale output: `--reset-cache / expo start --clear`.
- Inline requires** rewrite require to the first use site, so a module's top-level code runs on first use — cheaper start-up; side-effect order changes.
- Serialiser:** numeric module ids (`createModuleIdFactory`), `getModulesRunBeforeMainModule` (RN's `InitializeCore`), `getPolyfills`. Metro 0.87 (RN 0.87): source maps 2× faster, half the memory.
- Source maps:** dev maps are served beside the bundle (DevTools uses them); release `--sourcemap-output` + compose with hermes's map before upload.

Fast Refresh — the rules

- File exports **only components** → only that module re-runs; hooks' state kept.
- Exports anything else → that module *and its importers* re-run.
- Imported by code **outside the React tree** → **full reload**.
- `useState/useRef` keep values while hook order is unchanged; `useEffect/useMemo/useCallback` **always re-run** (deps ignored). Class state is not kept. `// @refresh reset` forces a remount.

Code splitting — the limits

Metro emits **one bundle** per app; `import()` resolves inside it. Dev servers can fetch lazily (*unverified*). Real split chunks on native: **Re.Pack** (Rspack/webpack, Module Federation). Expo Router async routes split on **web**. Hermes mmmaps bytecode, so a big bundle costs less than it seems.

Expo's Metro config adds

`EXPO_PUBLIC_*` inlining, `tsconfig` path aliases, `require.context` (Expo Router), CSS + web bundling, asset hashing, and production **tree shaking** (SDK 52+, `EXPO_UNSTABLE_TREE_SHAKING=1` with `EXPO_UNSTABLE_METRO_OPTIMIZE_GRAPH=1`). Always extend `expo/metro-config`, never replace it.

Example — metro.config.js for a monorepo

```
const { getDefaultConfig, mergeConfig } =
  require('@react-native/metro-config'); // Expo: 'expo/metro-config'
const path = require('path');
const root = path.resolve(__dirname, '..');
module.exports = mergeConfig(getDefaultConfig(__dirname), {
  watchFolders: [root], // packages/* live outside the app
  resolver: {
    nodeModulesPaths: [path.resolve(__dirname, 'node_modules'),
      path.resolve(root, 'node_modules')],
    blacklist: [/\/fixtures\/.*/], // never crawl these
    resolveRequest: (ctx, name, platform) =>
      ctx.resolveRequest(ctx, name === 'lodash' ? 'lodash-es' : name, platform),
  },
});
```

Monorepos and symlinks

- Symlinks resolve by default since RN 0.73 — but only inside `watchFolders`; the workspace root must be watched or “*Unable to resolve module*”.
- Two copies of `react` (app + hoisted) → “*Invalid hook call*”: force one via `nodeModulesPaths`, `extraNodeModules` or `disableHierarchicalLookup`.
- Expo's `getDefaultConfig` configures workspaces for you; start there.

Traps

- ! Works on the Mac, fails in Linux CI: `import './button'` vs `Button.tsx` — case-insensitive APFS hid it.
- ! Release app shows old JS: Release embeds the bundle at build; only Debug loads from Metro.
- ! “*EMFILE: too many open files*” / slow start → install Watchman.
- ! Changed an `.env` value, app still old — inlined at transform time: clear the cache.

Symptom → fix

Android device: no bundle	adb reverse tcp:8081 tcp:8081 (USB) or same Wi-Fi + dev-menu host
port 8081 busy	start --port 8082 — and reverse that port
Watchman “recrawl” warning	watchman watch-del <root> then watch-project <root>
new native dep, JS says “not found”	Metro cannot fix native: rebuild the app (pods / Gradle)

Remember

“Resolve, transform, serialise — cache per file, one bundle, deltas in dev.”